Project Progress - Week 3 of March 2025

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This is a file that contains the tracking of the activities related to the OFC project during the third week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the third week of March 2025. The key achievements include:

- The management of Python modules and compilation of LaTeX projects has been learned.
- A first approach to C++ modular development of the functionalities has been initiated.
- The reading of Draine's paper on radiative corrections in polarizability [1] has been initiated.
- The difficulties in implementing C++ modules have been discussed with Raúl. And intuition on how to proceed has been provided.
- The need to use the Discrete Dipole Approximation (DDA) for the calculation of optical forces for particles of size comparable to the wavelength has been discussed.

So far, the project is developing in some areas:

- The development of the algorithm to obtain the polarizability of the particles.
- The modular development, including makefile documentation and Raul's cpp repository and notes analysis.
- The development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.

2 Potential Tasks for the Week

The following tasks are proposed for the week of March 10th to March 15th, 2025:

- Exploration of CMakefile, include directory, and modular development (Notion)

- Develop the algorithm to obtain the polarizability of the particles.
- Implement the calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- Documentation of quasiestatic polarizability theory.
- Continue reading Draine's paper on radiative corrections in polarizability [1].
- Analysys of Raúl's `cpp-template-project` repository.
- Check of Raúl's class notes.
- Discuss the progress and difficulties with Raúl.

3 Week Progress

3.1 Monday, March 17th, 2025

The tasks for the week have been defined and the progress of the project has been analyzed. CMakeFiles and include directories have been explored, creating an initial structure (it doesn't compile yet) and notion pages for registering information about the template project `cpp-template-project`.

3.2 Tuesday, March 18th, 2025

The compilation issues with the CMakeFiles have been solved and the environment dependencies had been installed. The first test for optical forces calculation function has been implemented and tested. The function is able to calculate the optical forces given a medium permittivity, a set of dipolar momenta and a set of electric field gradients. A library for dielectric coefficients has been created. In it, a class for permittivities has been defined. A method for gold particles testing has been implemented, but yet not tested correctly.

4 Next Steps

The next steps for the project are:

- Check the wavelenght dependence of gold's permittivity.
- Explore compilator debbuger issues in VSCode.
- Check of Raúl's class notes.
- Continue reading Draine's paper on radiative corrections in polarizability [1].
- Continue with the development of the algorithm to obtain the polarizability of the particles.
- Implement the calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- Documentation of quasiestatic polarizability theory.

References

[1] B. T. Draine. "The Discrete-Dipole Approximation and Its Application to Interstellar Graphite Grains". In: *The Astrophysical Journal* 333 (Oct. 1988), p. 848. DOI: 10.1086/166795.